

Task requirement Technology Research & Innovation Engineer Position

Document + Web Chat Interface (FastAPI Backend)

1. Goal

Develop a backend service that:

- Accepts **PDF** and **DOCX** file uploads
- Enables conversational chat with follow-up queries based on uploaded content
- Falls back to web search when relevant document data is unavailable
- Uses **Redis** for caching responses
- Clearly labels responses as "**document**" or "**web**"

2. Document Processing

The system should support efficient ingestion and preparation of documents:

- Accept `.pdf` and `.docx` file formats
- Extract text and perform cleaning/normalization
- Split extracted text into smaller chunks for optimized retrieval
- Store processed content along with metadata:
 - File name
 - File ID
 - Timestamp
- Prepare searchable representations (e.g., embeddings or indexed text)

3. Chat Handling

To ensure contextual and meaningful conversations:

- Maintain conversation state per session using a **session ID**
- Store previous messages to preserve context
- Enable follow-up queries by combining:
 - Past conversation history
 - Retrieved document data

4. Query Flow

The system should process user queries using the following pipeline:

Step 1: Receive Request

- Accept user query along with session ID

Step 2: Cache Check (Redis)

- If a cached response exists:
 - Return cached result immediately

Step 3: Document Retrieval

- If no cache is found:
 - Search for relevant content in uploaded documents
 - Evaluate relevance of retrieved content

Step 4: Response Generation

- If relevant document data is found:
 - Generate response using document content
- If not relevant:
 - Perform web search and generate response using external data

Step 5: Source Labeling

- Attach a source tag to the response:
 - "document" → if derived from uploaded files
 - "web" → if derived from web search

Step 6: Cache Storage

- Store the generated response in **Redis** for future reuse

Step 7: Return Response

- Send the final response back to the user

5. Expected Outcome

The system should:

- Provide fast, context-aware answers
- Efficiently reuse cached responses
- Seamlessly combine document intelligence with web knowledge
- Clearly indicate the source of information for transparency